Anup Sathya

PhD Student, University of Chicago   anup@uchicago.edu   anupsathya.com

Research

Anup's research explores how to protect the home from the encroachment of addictive platforms in the attention economy. Believing that learning to calmly coexist with smartphones is one of the defining challenges of the next century, he designs, builds, and studies physical devices that shield us from the constant digital onslaught.

Research Positions

Pre-Doctoral Researcher — 2021 – 2022
Dept. of Computer Science, University of Maryland, College Park

Education

PhD, Computer Science — 2022 – present
University of Chicago
Supervisor: Dr. Ken Nakagaki | Thesis: in progress

MS, Computer Science — 2019 – 2021
University of Maryland, College Park
Supervisor: Dr. Huaishu Peng | Thesis: Enabling On-body Computing Using a Track Based Wearable

BTech, Electronics & Communication Engineering — 2014 – 2018
PES University, Bangalore
Supervisor: Suresh Padmanabhan | Thesis: Realtime On-chip Wireless Waveform Monitoring

Major Grants & Funding

Arts, Science + Culture Graduate Collaboration Grant ($3000) — 2024
University of Chicago + School of the Art Institute of Chicago

Crerar Fellowship ($5000) — 2022
Dept. of Computer Science, University of Chicago

Daniels Fellowship ($5000) — 2022
Dept. of Computer Science, University of Chicago

Full tuition remission (~$56k) — 2019 – 2021
University of Maryland, College Park

Awards & Honors

Linda Tischler Award — Recognizing Emerging Design Talent 2025
Fast Company

Innovation by Design Award (for Attention Receipts) 2025
Fast Company

Lifestyle Accessories Award — Runner Up (for Attention Receipts) 2025
Core77

Creative Hack Award Finalist (for Attention Receipts) 2025
WIRED Japan

Special Recognition for Outstanding Reviews (11x) 2023 – 2026
DIS 2023, CHI 2024, CHI 2026 (3×), DIS 2026 (5×), Creativity & Cognition 2026

Design Awards — Tools Award (for Fibercuit) 2023
Core77

Nominated for Graduate Assistant of the Year (top 2%) 2020
University of Maryland, College Park

Zonal Winner, National Robotics Championship 2016
IIT Bombay, India

Publications

2026

[17] ACM CHI '26 — Full Paper
Attention Nooks: Situated Frictions to Foster Intentional Technology Use
Anup Sathya, Ken Nakagaki

[16] ACM TEI '26 — Full Paper
PopTuber: Discretely Reshaping High Resolution 3D Curves with "Serial Multistable" Tubes
Willa Yunqi Yang, Yifan Zou, *Anup Sathya*, Ken Nakagaki
Best Demo Award (People's Choice)

[15] ACM TEI '26 — Work-in-Progress
FocuShift: Actuated Intervention to Prevent Smartphone Overuse via Phone-Attached Shape-Changing Device
Harrison Dong, *Anup Sathya*, Ken Nakagaki

[14] ACM TEI '26 — Studio
Attention-Preserving Tangible Interfaces: Using Physicality and Materiality to Preserve Attention in the Home
Anup Sathya, Ken Nakagaki

2025

[13] ACM UIST '25 — Full Paper
Shape n' Swarm: Hands-on, Shape-aware Generative Authoring with Swarm UI and LLMs
Matthew Jeung, *Anup Sathya*, Wanli Qian, Steven Arellano, Luke Jimenez, Ken Nakagaki
Best Demo Award

[12] ACM UIST '25 — Workshop
Facilitating Longitudinal Interaction Studies of AI Systems

Tianyi Long, Sihan Wang, Émilie Fabre, Tingying Wang, *Anup Sathya*, Jing Wu, Savvas D. Petridis, Danqing Li, others

[11] ACM DIS '25 — Full Paper
Cybernetic Marionette: Channeling Collective Agency Through a Wearable Robot in a Live Dancer-Robot Duet
Anup Sathya, Jiasheng Li, Zeyu Yan, Adriane Fang, Bill Kules, Jonathan Martin, Huaishu Peng

2024

[10] ACM UIST '24 — Full Paper
SHAPE-IT: Exploring Text-to-Shape-Display for Generative Shape-Changing Behaviors with LLMs
Wanli Qian, Chen Gao, *Anup Sathya*, Ryo Suzuki, Ken Nakagaki

[9] ACM UIST '24 — Full Paper
CARDinality: Interactive Card-shaped Robots with Locomotion and Haptics using Vibration
Aditya Retnanto*, Emilio Faracci*, *Anup Sathya**, Yu Kai Hung, Ken Nakagaki [*equal contribution]

[8] ACM CHI '24 — Full Paper
Attention Receipts: Utilizing the Materiality of Receipts to Improve Screen-time Reflection on YouTube
Anup Sathya, Ken Nakagaki
FastCompany Innovation by Design Award, Core77 Lifestyle Accessories Award, WIRED Japan Creative Hack Award Finalist

[7] ACM UIST '24 — Poster
Game Jam with CARDinality: A Case Study of Exploring Play-based Interactive Applications
Emilio Faracci, Aditya Retnanto, *Anup Sathya*, Ashlyn Sparrow, Ken Nakagaki

2022

[6] ACM UIST '22 — Full Paper
Fibercuit: Prototyping High-Resolution Flexible and Kirigami Circuits with a Fiber Laser Engraver
Zeyu Yan*, *Anup Sathya**, Sarah Yusuf, Jyh-Ming Lien, Huaishu Peng [*equal contribution]
Best Demo Award (People's Choice), Core77 Tools Award - Notable

[5] ACM IMWUT (UbiComp) '22 — Full Paper
Calico: Relocatable On-cloth Wearables with Fast, Reliable, and Precise Locomotion
Anup Sathya, Jiasheng Li, Tauhidur Rahman, Ge Gao, Huaishu Peng

[4] ACM UIST '22 — Demo
Demonstration of Fibercuit: Techniques to Prototype High-Resolution Flexible and Kirigami Circuits with a Fiber Laser Engraver
Zeyu Yan*, *Anup Sathya**, Saadat Yusuf, Jyh-Ming Lien, Huaishu Peng [*equal contribution]

2021

[3] ACM CHI EA '21 — Poster
Towards On-the-wall Tangible Interaction: Using Walls as Interactive, Dynamic, and Responsive User Interface
Zeyu Yan, *Anup Sathya*, Pedro Jorge Carvalho, Yongquan Owen Hu, Ang Li, Huaishu Peng

2018

[2] IEEE ICACCI '18 — Short Paper
Realtime On-chip Wireless Waveform Monitoring
Anup Sathya, S. Balaji, Abhinav Gupta, S. Padmanabhan

2017

[1] IEEE TENCON '17 — Short Paper

Visual positioning system for automated indoor/outdoor navigation
Anup Sathya, Abhinav Goel, S. Padmanabhan

Research Dissemination

Press

University of Chicago Magazine, 'Fit to print' — about *Attention Receipts* 2025

Hackster.io, 'A Game Changer' — about *CARDinality* 2024

inews UK, opinion piece citing *Attention Receipts* 2024

404 Media, 'Would You Waste Less of Your Life Online If You Got Daily Attention Receipts?' 2024

Adafruit Blog, 'Calico: A Wearable Robotic Assistant #WearableWednesday' 2023

Indian Express, 'This is Calico, a tiny robot that can be your dance instructor, workout tracker, and more' 2023

Communications of the ACM, 'A Wearable Robotic Assistant That's All Over You' 2023

IEEE Spectrum, 'A Wearable Robotic Assistant That's All Over You' 2023

The Verge, 'Wearable robot, why not?' 2023

Maryland Today, 'Where-able sensors' 2022

all3dp, Hackaday (×2), Hackster.io — coverage of Fibercuit and Calico 2022

Podcasts & Interviews

CBC Radio, 'As It Happens' — interviewed about *Attention Receipts* 2024

Put on Your Best...Robot, 'Over Coffee' — interviewed about *Calico* 2024

Hackster Café — interviewed with Huaishu Peng and Zeyu Yan about *Calico* 2022

Invited Talks

Intentional HCI — Designing for Intentional Technology Use 2025
University of British Columbia, Advanced Methods for Human-Computer Interaction, Dept. of Computer Science

Teaching Experience

Teaching Assistant, A Practice in Art & Technology 2024
University of Chicago, Dept. of Computer Science | Instructor: Ken Nakagaki

Teaching Assistant, Intro to Human-Computer Interaction 2023
University of Chicago, Dept. of Computer Science | Instructor: Ken Nakagaki

Teaching Assistant, Actuated User Interfaces and Technologies 2022
University of Chicago, Dept. of Computer Science | Instructor: Ken Nakagaki

Service

Web Chair 2026
ACM TEI 2026

Organizer, People and Tech Seminar 2024
Dept. of Computer Science, University of Chicago | 2 quarters

Reviewer 2023 – present
CHI 2023 (3×), DIS 2023 (1×), CHI 2024 (1×), DIS 2024 (2×), CHI 2025 (6×), DIS 2025 (1×), TEI 2025 (2×), UIST 2025 (2×), CHI 2026 (5×), Creativity & Cognition 2026 (1×), DIS 2026 (7×), IMWUT 2026 (1×), NordiCHI 2026 (1×), TEI 2026 (1×), UIST 2026 (1×)